

Using a Duplex Time-to-Digital Converter for Metastability Characterization of an FPGA

Florian Huemer*, Thomas Polzer† and Andreas Steininger*

*TU Wien, Institute of Computer Engineering, Vienna, Austria

{fhuemer,steininger}@ecs.tuwien.ac.at

†UAS Technikum Wien, Department of Electronics and Telecommunications, Vienna, Austria

thomas.polzer@technikum-wien.at

Abstract—In view of the increasing number of clock domains found in modern ASICs, the precise characterization of metastability at their boundaries becomes crucial. In some cases, the conventional approach does not provide a sufficient level of detail information. As an alternative approach, the use of a time-to-digital converter based on a tapped delay line has been proposed. In this paper we extend the latter by an additional tapped delay line thus allowing to further refine the concept. We present the underlying concept, its implementation, as well as experimental measurements on an FPGA platform that reveals significant variations in the metastable behavior of different FPGA boards of the same type.

I. INTRODUCTION

As a consequence of the rapid growth of system complexities, the number of clock domains in a system grows as well. For appropriate interaction of and communication between function blocks across clock domain boundaries, synchronizers need to be employed. As it is well known that perfect (i.e. completely upset-free) synchronization is in general impossible, the best a synchronizer can do is reduce the probability of a metastable upset to a negligible value. This, however, comes at the price of a performance penalty, and therefore a balance between reliability of a synchronizer and the delay incurred by it must be made in the design. For the common case of uncorrelated clocks this is governed by the following equation [1]:

$$MTBU = \frac{1}{f_{clk} \cdot \lambda_{dat} \cdot T_0} e^{\frac{t_{res}}{\tau}} \quad (1)$$

Here f_{clk} is the clock rate of the receiving domain, connected to the clock input of the receiver flip flop, and λ_{dat} the rate of data transitions (aligned to the sending domain's clock), connected to the same flip flop's data input. With uncorrelated transitions on its clock and data inputs, the flip flop will occasionally experience setup/hold violations, in which case its output may remain at an undecided voltage between a clean HI and a clean LO, i.e. in a metastable state. This state will eventually resolve, but the time this takes depends on how deep the flip flop has been driven into metastability, and is essentially unbounded. The resolution time t_{res} is the time allowed by the designer for that metastable state to resolve, before the output is consumed by the subsequent flip flop. It is roughly equivalent to the time penalty the synchronizer introduces. As can be seen in Equ. 1, increasing t_{res} results

in an exponential increase of the MTBU, which expresses the mean time between upsets, i.e. the inverse of the upset rate.

This basic relation between the design parameters is quantitatively characterized by the two technology-related parameters T_0 and τ , the latter being called the resolution time constant. Obviously, to tune the resolution time for a targeted MTBU, the designer must know T_0 and τ . Unfortunately, it is relatively challenging to determine these values sufficiently precise, as they depend on technology, routing and load and change with process variations, voltage and temperature.

There are established methods for metastability characterization (i.e. determination of T_0 and τ), but these reach their limits when (a) high precision is required and (b) the relatively simplistic model underlying Equ. 1 does not fully apply, and other effects like oscillation enter the picture. Unfortunately, modern process technologies rapidly approach these limits: With hundreds of synchronizers on a single ASIC, an overly conservative choice of the MTBU cannot be afforded, as the resulting performance penalty would become prohibitive. Thus *precisely* knowing T_0 and τ becomes mandatory. Furthermore, with RC constants (inertial delays) being optimized further and further, pure (transport) delay is getting noticeable, and in this setting oscillation may occur.

With this motivation we propose and investigate a method for a refined metastability characterization of flip flops¹. Its principle of using a time-to-digital converter (TDC) is substantially different from the conventional approach and promises better results. While such a TDC based approach has already been introduced in [2], we further improve its implementation concept by introducing a parallel TDC structure.

The rest of the paper is structured as follows: In Section II we will briefly sketch the problem of metastability characterization and introduce the conventional characterization method, as well as related work. Next, in Section III we will present our new double-TDC concept, whose evaluation through a comprehensive experimental study will be the focus of Section IV. The paper will be concluded with Section V.

II. BACKGROUND AND RELATED WORK

A. Basic Metastability Measurement

In the analog domain a metastable storage element exhibits an “intermediate” output voltage V_{meta} that lingers between

¹This research was partially supported by the Austrian Science Fund (FWF) project SIC (grant P26436-N30).

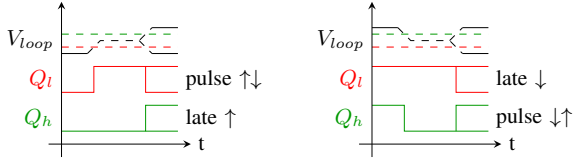


Fig. 1: Possible metastability behaviors of a storage loop; outputs of low- (Q_l) and high-threshold buffers (Q_h)

the thresholds for well-defined HI and LO logic value. However, this only applies for the output voltage V_{loop} of the actual storage primitive (cross-coupled inverter pair, e.g.), which is always followed by a buffer primitive (actually an inverter, but here we use a buffer to simplify the explanation) for decoupling, before being provided as a (flip flop or latch) cell output Q . That buffer typically does not convey the intermediate voltage as such, but rather produces some type of “digital output”, depending on its threshold V_{th} . As shown in Figure 1 for the case of $V_{th} > V_{meta}$ (high-threshold, green trace) there are the following possible reactions:

Case A – V_{loop} initially LO: The metastability raises V_{loop} to a value of V_{meta} , but since $V_{th} > V_{meta}$, the output Q remains LO until the metastability resolves. In case it finally resolves to LO, we do not see any reaction at Q , the metastability has been masked. Should it resolve to HI we see a clean transition, which is late, as it occurs only after metastability has resolved.

Case B – V_{loop} initially HI: When going down from HI to V_{meta} , V_{loop} crosses V_{th} and immediately causes a falling transition at Q . In case the metastability finally resolves to LO, V_{loop} continues to fall without any threshold crossing, so overall we see a clean transition with nominal delay. However, should the metastability resolve to HI, we see another transition at Q . This means we have experienced a negative glitch whose width equals the duration of the metastable state.

In a similar fashion, the possible behaviors of a low-threshold buffer ($V_{th} < V_{meta}$) can be deduced: no transition, regular transition, late transition, or positive glitch.

Traditionally, metastability measurement on the cell level is concerned with the case of late transitions only. In more detail a characterization is performed as follows: The unit under test is provided with uncorrelated transitions on data and clock, which will occasionally drive it into metastability. Its output is observed a given delay Δ after the clock edge and checked for metastability by comparing the observed logic value with a reference that is obtained after a much larger delay (i.e. in the resolved state). In case of a mismatch, a metastable upset is assumed and a counter increased. After a certain measurement interval T_m the count is read and related to T_m , thus obtaining the MTBU. Then the experiment is repeated for a different setting of Δ – which obviously corresponds to the resolution time t_{res} – and again the MTBU determined. By collecting several such points the function $MTBU(t_{res})$ can be obtained, which is, as can be seen from Equ. 1, a straight line when drawn on a semilogarithmic scale. In this function, τ determines the slope, while the offset depends on T_0 .

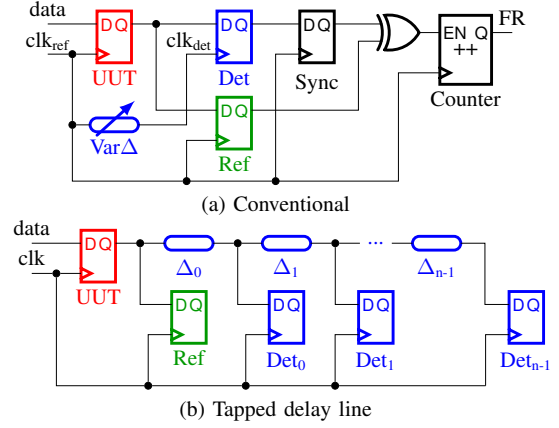


Fig. 2: Basic circuit for metastability characterization

Figure 2a shows the principle of the circuit that is conventionally used. The flip flop termed *UUT* is the unit under test; its output is observed by the detector flip flop *Det* after a variable delay $Var\Delta$. After synchronizing *Det*'s output through *Sync*, it is compared with the reference that is obtained by *Ref* a full clock period later (which is assumed to be sufficient resolution time even for very deep metastability). If the XOR indicates a mismatch, the counter is enabled and incremented by one.

The tricky part in this experiment is the implementation of $Var\Delta$: For contemporary technology (like 28nm, e.g.) a metastable state lasting for more than 1ns will be a very rare event, and, as evidenced by the exponential dependence of MTBU on t_{res} in Equ. 1, the probability of observing even longer durations rapidly decreases. Hence, to obtain a graph with reasonable quality, a controllable delay is required with a resolution (and accuracy) in the 10-picosecond range.

In the literature, a variety of different approaches for implementing such a delay can be found (Due to space restrictions we can only present a restricted selection here, for a more thorough overview, see [2]). Some of these, however, cannot be applied for our purpose: Starved buffers as proposed in [3], e.g., are not available on an FPGA platform which we want to use as our target. The use of a precise external pulse generator, as reported in [4] back in 1996, suffers from timing uncertainties in the cabling that are no more acceptable today. By triggering the flip flop under test with the rising clock edge, while sampling its output with the falling one, [5], [6], [7] utilize the HI-pulse width of the clock for determining the resolution time. This approach scales badly for recent technologies, as the appropriate control and variation of clock pulse width in the ps-range is difficult to accomplish. In [8] the authors propose the use of an inverter chain with fixed delay, and they vary the clock frequency to obtain different resolution times. This, however, also changes f_{clk} in addition to t_{res} in Equ. 1, which makes the extraction of τ more cumbersome.

The use of a digital clock manager [9] or PLL [10] has proven a viable approach on an FPGA. In conjunction with careful routing this yields a useful late transition detector

with decent temporal resolution. However, this is exactly the “conventional” approach we want to surpass with respect to efficiency of the measurement process and quality of data.

The tapped chain of logic functions from [11], the tapped inverter chain from [12], and apparently also the “delay line” mentioned in [10] are very close to our proposed approach in the way they implement a delay. However, our method here is based on a fundamentally different principle that was first presented by [2]. Rather than implementing a *variable* delay for defining a “borderline resolution time” and just measuring whether the observed metastability lasted longer or not (binary), as in the conventional approach, it utilizes a *fixed* tapped delay line to implement a time-to-digital converter that directly measures the actual resolution time (discretized analog). This underlying principle will be outlined in the subsequent subsection.

B. TDC-based Metastability Measurement

The key component of the late transition detector (LTD) proposed in [2] is a tapped delay line (TDL) implemented using the carry logic of the FPGA fabric of the Xilinx Zynq-7000 SoC on a ZedBoard development kit.

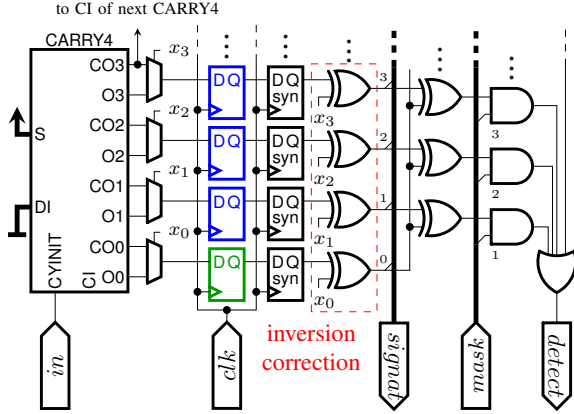


Fig. 3: Circuit diagram of the tapped delay line (from [2])

Figure 3 shows the implementation details of the design. The marked flip flops are colored to indicate their role in the circuit (see Figure 2b for comparison). The TDL itself, is implemented by a number of 4-bit carry chain segments (CARRY4 elements) serially aligned to form one n-bit carry chain. Specifically in an FPGA environment the carry chain makes an attractive delay element, as, according to its purpose, it is optimized for low delay and short routing paths, thus yielding low dependence on the routing and good temporal resolution. The output of the UUT (not shown in the figure) is connected to the *in* port of the circuit. The idea is to sample the individual taps of the TDL after a given time (here one full clock period) and observe up to which tap the transition of the UUT output made it. Clearly, in case of metastability the late transition will not get as far as in the nominal case. So the “signature” of a given transition (i.e. the bit pattern it produces

at the TDL tap outputs) gives an indication on the transition timing. This signature is passed through a synchronizer array to remove potential metastable values it might contain (note that at this point one is only interested in a clean digital signature indicating the delay imposed by metastability of the UUT, rather than observing a potential metastability in one of the TDL’s output flip flops). The available CARRY4 elements provide two types of output, namely *O* and *CO*, selectable by the configuration bit vector $x_0 \dots x_{n-1}$ (in the design’s bitstream). In [2] the authors use an alternating pattern for the selection of the *O* and *CO* outputs [13] to ensure that the taps are more evenly spaced. For the taps where the *O* output has been selected an additional inversion is required to correct a previous inversion caused by the internal structure of the CARRY4 element. This is accomplished in the box termed *inversion correction*.

Tap 0 of the TDL (the green one at the bottom of the chain) has the lowest delay (measured from the output of the UUT), thus allowing for the longest resolution time. So it is used as a reference for the desired final output, allowing to check for the occurrence of metastability: If the signature contains any bit that does not match the reference, it is representing a late transition and forwarded to the analysis (output *signat*). This is triggered by the signal *detect* that is derived from an array of XOR gates (doing the comparison with tap 0) and an array of AND gates controlled by signal *mask* (allowing to reduce the sensitivity of the detector). Additionally a counter is used to generate time stamps that allow to monitor the temporal spacing of detection events.

A key advantage of the approach from [2] is that it provides way more detailed information about the UUT behavior: Recall that in the conventional approach the only available information is the number of occurrences for which the UUT’s output delay was larger than the selected resolution time. In contrast, in [2] no choice for the resolution time must be made, the circuit returns, for every metastable event, the actual output delay, and it adds a time stamp for the event. With this information available on a host PC, one can, among other things, study the effect of different choices for the resolution time, or investigate the temporal distribution of upsets. This comes at the price of high demands on the communication interface between measurement platform (FPGA board) and host PC. The authors propose a number of provisions to reduce the bandwidth requirements, such as masking of extremely frequent events to allow the targeted collection of rare ones, or special encoding of the transmitted data.

III. IMPROVED TDC-BASED LTD

The existing TDC-based measurement circuit discussed in the previous section has some shortcomings. Due to different propagation speeds for rising and falling edges, some (short) pulses can get “filtered” by the delay line (i.e. one edge overtakes the other, annihilating the pulse). In the results of [2] this manifested itself in a flat region in the *upset rate* plot. To address this issue our improved measurement circuit employs two TDLs, where one gets the original output of the

UUT while the other one gets an the inverted version. Since the propagation speeds of rising and falling edges should be similar in both delay lines a pulse that is shortened in one delay line should experience the exact opposite transformation in the other one (i.e. pulse broadening). Hence, pulses that are completely annihilated in one delay line should still be visible in the other one.

A. Circuit Details

An overview of our approach is shown in Figure 4. The blocks labeled TDL A and B are basically instances of the circuit shown in Figure 3. The input of the UUT is fed by a (divided) clock signal, which must be uncorrelated to the clock of the UUT. The frequency of this clock can be controlled by the *div* input, which allows to control the upset rate of the UUT. The *cal_{en}* input is used to feed the TDL inputs with a calibration signal. Depending on *cal_{sel}* this signal can either be the undivided *data* clock or some divided version of it.

The *detect* outputs of the TDL are then fed into a block that decides, based on the current measurement mode (one delay line or both) and some user configurable temporal masking (useful for certain calibration runs), whether the event is forwarded for transmission to the host PC. Moreover a 16 bit counter value (*cnt*) is provided as a time stamp for the event. Similar to [2] we also use 512 taps for our delay lines. Hence, for each detected event 1040 bits are written to the Event FIFO. In the following processing steps (right part of the figure) this data is first compressed to 256 bits per event and then transferred to the processor core of the SoC via an AXI streaming interface. Subsequently a Gigabit Ethernet link is used to transfer the data to a host PC.

A crucial aspect of the implementation of this circuit is the placement of the delay lines relative to the UUT and the multiplexers at their inputs, which makes manual placement of these components essential. To reduce the delay between UUT and TDL inputs, the UUT and the TDL input multiplexers are placed in close proximity to the inputs of the delay lines. Both TDL input multiplexers can be placed into the same LUT to exactly control the point where the input signal is forked (in this FPGA technology a LUT can have up to 2 outputs). Still, different synthesis runs may yield slightly different routing paths and hence delays. These uncertainties must be taken into account during measurement and calibration.

Notice that the UUT has a dedicated clock input (*clk_{UUT}*). To be able to trigger a sufficiently large number of upsets in the slave latch of the UUT in a reasonable amount of time, it must be possible to reduce the high time of its clock signal. For our measurements we use a PLL to generate a clock with 8.3% duty cycle out of the 125MHz *clk* input. For the *data* input, we use a (uncorrelated) 200MHz signal.

B. Calibration

Every pair (T_i, T_{i+1}) of neighboring taps defines the borders of a bin B_i , and a transition reaching T_i but not T_{i+1} is associated to bin B_i which corresponds to a certain (interval of) metastable UUT output delay(s). However, since the

delays of the individual taps are usually subject to significant variations, the bins have different sizes and the TDL must be calibrated [11]. The bin size needs to be precisely known, as it affects the slope of the $UR(t_{res})$ graph, from which the most important parameter, namely τ is determined.

The main sources of inaccuracy are routing asymmetries (inside and between CARRY4 elements), process variations, as well as temperature and voltage dependence of gate and routing delays. As already mentioned above, in addition rising and falling edges have different propagation speeds along the chain. In [2] even non-monotonic behavior was observed, i.e., an edge appears earlier on a tap T_{i+1} than on T_i . In principle, the static effects can be eliminated by a calibration run, while the dynamic ones are more difficult to handle. In the approach from [2], one calibration run is performed before and one after each measurement run. It turns out that the discrepancies between these two calibration results are relatively small, which gives evidence that the dynamic variations are not too significant. We use the exact same techniques for our circuit and calibrate each delay line individually.

The following process is used to mitigate these substantial static uncertainties:

In a first phase, the relative tap order is identified. To this end, the TDL is fed by a divided version of the *data* clock (using the calibration multiplexer in Figure 4), and the signatures produced by the TDL are recorded. In particular, within a guard band around a transition of the input signal, it can be observed that some taps perceive the transition earlier than expected and others later. This allows establishing a partial order of the taps, which, after collection of sufficient data, can be transformed into a global relative order (wherein pairs of taps that showed inconsistent partial order are grouped into a single logical tap, for details see the original paper).

In a second phase, the relative bin size is determined by comparing the counts observed for the individual bins. Given that the calibration clock is uncorrelated with the sampling clock of the TDL, one can expect to see (statistically) equal counts in all bins. Hence mismatches directly indicate discrepancies in the bin sizes and can be used for calibration – the count directly yields the respective relative bin size.

Finally, the absolute bin size is calibrated in a third phase. Here the frequency of the calibration clock is increased to the point where its half period is shorter than the overall delay of the TDL, thus obtaining two transitions in each signature. Clearly, the sum of bin sizes enclosed between these two transitions must equal the half period, which is known. So the relative bin sizes determined in the previous phase can be mapped to absolute time. For further details, we refer to [2].

IV. RESULTS

A. Verification Measurement

The verification measurement was designed to recreate the results presented in [2]. Therefore, the obtained upset rate curve ($UR = 1/MTBU$) was compared to the one from the latter paper. To our surprise, the results did not match, even though we had taken care to use the same type of board with

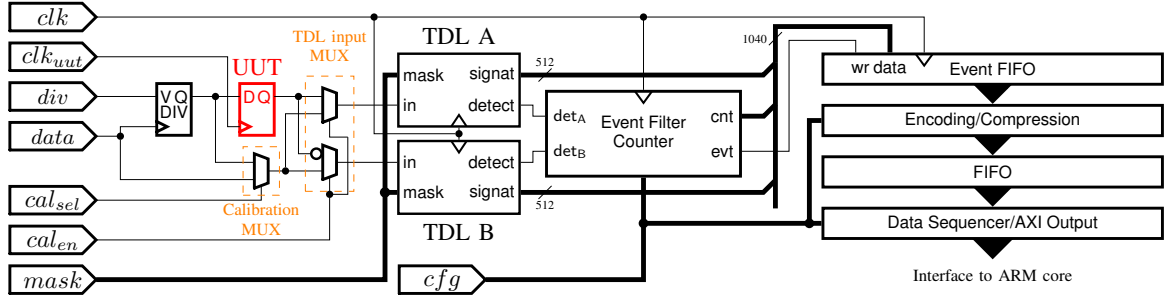


Fig. 4: Improved TDC-based late transition detection with two delay lines

the same type of FPGA. Specifically, no double pulses were registered after a measurement time of more than two days. After five days of measurement, at least some 0101 double pulses became visible (see Figure 5a).

We therefore decided to extend our measurements to other boards of the same type. One board (Board 2) showed pulses of negative polarity (010), which is the exact opposite pulse direction of the first board (see Figure 5b), while another board (Board 3) was found to have the same characteristic as the one from [2] (see Figure 5c).

These results emphasize that even boards of the same type and exactly the same FPGA may exhibit fundamentally different metastability behavior. It is therefore safe to conclude that the variability between the boards is significant and each board must be tested and calibrated separately to get an optimal metastability resolution behavior. To summarize these basic measurements, Table I presents the measured metastability resolution constants (τ) for the three presented boards.

TABLE I: Metastability resolution constants τ [ps]

Board	Polarity	Master	Slave	Double Pulse
Board 1	rising	68	310	-
	falling	76	315	-
Board 2	rising	62	230	-
	falling	75	250	-
Board 3	rising	63	180	460
	falling	77	348	-

B. Delay Line Differences

As already stated before, delay line A works non-inverted (as in [2]), while the second delay line inverts the signal at its input. This allows us to investigate the propagation delays of the different transition polarities, and how their mismatch affects the pulse size and slope of the measured curves. For analysis, we compared the failure rate plots captured by the two delay lines A and B. Figure 6 shows the result for Board 2; the other results are similar. Please note that the inversion of delay line B has been corrected in the figure.

Figure 6 clearly shows that the failure rate curves for rising and falling edges, as well as for pulses have a relative offset, while the slope is identical, indicating that τ is not distorted. Furthermore, the maximum of the 010 pulses for the delay line A curve shows a wide plateau, while it is much more rounded

for delay line B. To understand this behavior, we performed an additional analysis, namely the measurement of transition offsets between the two delay lines.

C. Transition Propagation Behavior

To better understand the phenomena described in the last section, we analyzed the correlation between corresponding transitions in both delay lines. As our circuit records both delay lines in tandem, we have a correlation between each measured signal on delay line A and B. Using this correlation, we can measure how far the edge traveled within the delay line and assign it a corresponding time stamp (t_A , t_B). We then subtract these time stamps and associate the resulting Δ with the polarity of the edge as well as the time stamp of delay line A as a reference. By averaging over all Δ for a given polarity and reference time stamp, we obtain a $\bar{\Delta}$, which is a measure of the delay difference between the two delay lines. The result is shown in Figure 7.

As the figure shows, for a rising edge on delay line A (recall that the corresponding edge on delay line B would be a falling edge) the delay difference Δ is nearly zero, while for the opposite polarity, the delay difference is around 180ps. The only deviation from these values is near 6ns. This is the area where the first pulses start to form (due to the shifted timescale, this is the region around 700ps in Figure 6).

These differences in propagation delay on the two delay lines indicate a pulse shortening on one and a pulse broadening on the other delay line (see Figure 8), explaining the differently formed maximum area for 010 pulses as well as the relative curve offset in Figure 6. Furthermore, the nearly constant value of Δ indicates a good calibration result as well as a uniform delay behavior of our delay line.

The only area we are not able to analyze with this circuit is the connection between the UUT and the fork at the input of the delay line (TDL input multiplexer in Figure 4). As the signal propagates in only one polarity in this area, we cannot measure any delay differences. Nevertheless, since this connection is by construction very short, the error imposed is very small.

V. CONCLUSION

In our attempt to increase the precision of metastability characterization of FPGAs we have adopted the method based

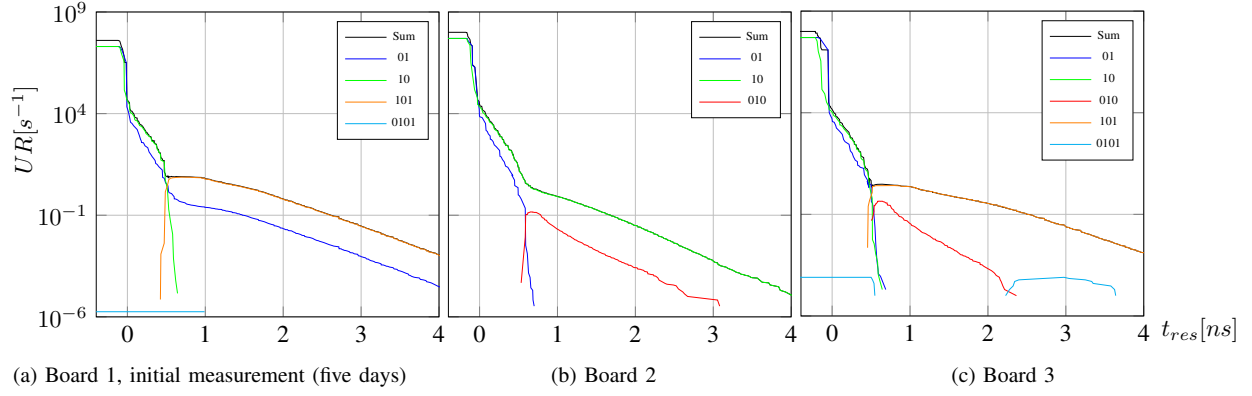


Fig. 5: Measurement results

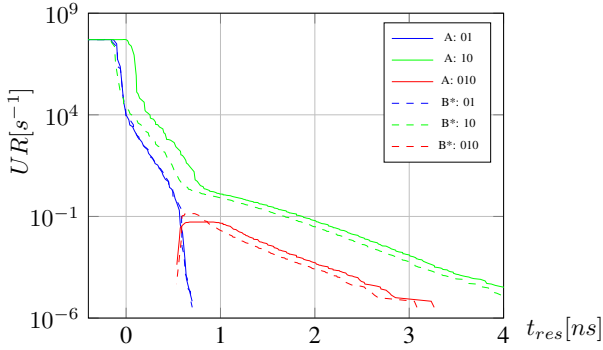


Fig. 6: Result comparison created from the two delay lines

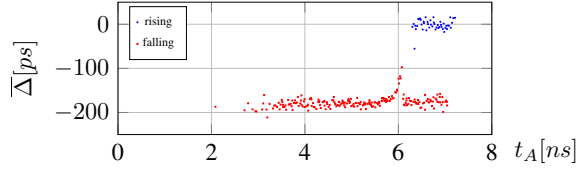


Fig. 7: Delay differences for transition of given polarity

on a time-to-digital converter and further refined it by using a second TDL in parallel. This allowed us to get a more faithful representation of glitches in the metastability resolution trajectory, which are not at all recognized by the traditional approaches, and partly filtered out in the single-TDL approach.

Our validation measurements on three FPGA boards of the same type not only confirmed this superior pulse recognition capability, they also revealed interesting differences in the metastability resolution behavior of the three boards. While

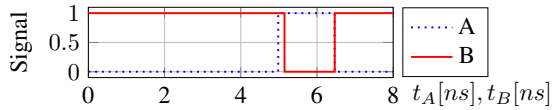


Fig. 8: Pulse as measured by both delay lines

each of the boards, as expected, exhibited glitches in the region of deep metastability, those glitches were exclusively 010 on one board, exclusively 101 on the other, and both glitch polarities were found on the third one. This seems to be due to different thresholds on the UUT output buffers.

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